Zach Greenberg

Mechanical Engineering Student Specializing in Mechatronics & Advanced Prototyping

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Education

Georgia Institute of Technology, Atlanta, GA — Expected Graduation: *May 2027, May 2028*

- Candidate for Bachelor of Science in Mechanical Engineering — GPA 3.84
- Candidate for Masters of Science in Mechanical Engineering — GPA 4.00

Experience

Assistant Project Manager Intern at Batchelor & Kimball, Atlanta GA — ***May 2026 – Present***

- Managed execution and mechanical coordination of industrial HVAC and cooling infrastructure for data centers
- Optimized cooling workflows and equipment layouts to maximize thermal efficiency in high density facilities
- Conducted system risk assessments and root cause analysis during complex piping and hardware installation
- Engineered a schedule tracking tool to identify critical path conflicts and accelerate equipment delivery

Automation and Controls Intern at Dennis Group, Salt Lake City, UT — ***May 2025 – August 2025***

- Developed Python and Studio 5000 logic enabling integration of third-party hardware with Allen-Bradley PLCs
- Contributed to electrical and controls design, writing PLC logic, and creating animated HMI interfaces in Ignition
- Developed and animated machine screens, debugged automation issues, and supported system commissioning

Invention Studio Casting Master, Atlanta, GA — Awarded Best New PI of Spring 25 — ***March 2025 – Present***

- Oversaw foundry operations, tooling procurement, equipment maintenance, and user safety training programs
- Facilitated operations in nation's largest student-run makerspace, managing 3D printers and CNC machines
- Trained and mentored 500+ students in hands-on fabrication, guiding varied projects from concept to execution
- Maintained metalworking and woodworking equipment, ensuring strict safety standards and tool calibration

Internship at Mobility Prosthetics, Murray, UT — ***June 2023 - August 2023, May 2024 – July 2024***

- Enhanced prosthetic fitting through refining and modeling 3D prints and managing timely print jobs
- Organized and managed a computer system to keep a record of placement, progress, and receipt of orders
- Created an organizational system for hardware to be sorted by patient name and type of equipment

Projects (see portfolio link for more detail)

Marine Robotics, Georgia Tech, Atlanta, GA — ***August 2023 - Present***

RoboSub – the creation of an autonomous underwater vehicle tasked with completing several challenges

- Engineered electromechanical manipulation and collection systems for an autonomous underwater vehicle
- Conducted pressure testing and FEA to ensure the structural integrity of sealed housings at operational depths
- Fabricated custom electronics housings and thermal management structures to protect critical internal systems

Medical Robotics, Georgia Tech, Atlanta GA — ***August 2023 – December 2024***

Humanoid Robotics – Spearheaded creation of a humanoid robotic hand for medical research

- Optimized finger kinematics and thumb opposition with rapid FDM/SLA prototyping and mechanical validation
- Integrated sub-miniature actuation systems within strict anatomical space constraints to ensure fluid motion
- Transitioned the final prototype to clinical testing phases, where it is currently undergoing amputee evaluation

Research

EPIC Labs, Georgia Tech | Mechanical Lead, Wearable Motion Capture Research — ***January 2026 - Present***

- Spearheaded mechanical design of wireless motion capture suits for prosthetic and exoskeleton testing
- Engineered ergonomic sensor enclosures and fabric closures utilizing SLA/FDM rapid prototyping
- Defined mechanical constraints to maintain sensor orientation and data integrity during dynamic movement
- Documented hardware iterations and validation protocols to prepare the platform for clinical subject trials

University of Utah, Neurosurgery Research Assistant, Salt Lake City, UT — **June 2023 – August 2023**

MEMBRANE Study: Endovascular Embolization Device Clinical Trial (TRUFILL®)

- Collected and recorded performance data in capillary structures to improve recovery and limit recurrence
- Analyzed and organized study data, procedures, and results from patient records and study booklets
- Streamlined the study's processes through the structuring and categorization of research files

Skills

Mechanical Design: DFM, **Kinematics,** FEA (Static/Thermal), GD&T, Mechanism Design, Thermal Management
CAD & Fabrication: SolidWorks, Fusion 360, 3D Printing (SLA/FDM), CNC Machining, Metalworking, Woodworking
System & Controls: Python, C++, MATLAB, Studio 5000 (PLC), Ignition (HMI), Arduino, Simulink, GIT/GitHub